

Constrained Geo-Semantic Identifiers and Utility-Supervised Dynamic Multi-Expert Retrieval for Next-POI Prediction

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Abstract—Next point-of-interest prediction requires selecting a destination from a large and heterogeneous POI catalog. We present GeoSemID, which augments atomic POI identities with constrained geographic, categorical, and semantic codes; uses utility-supervised dynamic multi-expert retrieval; and preserves retrieval evidence for candidate-constrained language-model ranking. Across 1,818 CA, 988 NYC, and 2,206 TKY test instances, GeoSemID achieves 42.48%, 44.65%, and 46.85% HR@10, respectively. Its dynamically fused top-50 candidate pool reaches target recall of 60.00%, 62.40%, and 64.20% on the three datasets.

Index Terms—Next point-of-interest prediction, trajectory modeling, semantic identifiers, multi-expert retrieval, dynamic routing, large language models.

I. INTRODUCTION

Next point-of-interest (POI) prediction is a fundamental problem for intelligent transportation systems, location-based services, and personalized mobility assistance. A useful predictor must reconcile persistent user preferences with immediate movement intent while accounting for temporal regularity, geographic reachability, and the functional meaning of locations. Although sequential and spatiotemporal models have substantially improved trajectory encoding, the decision space remains a large catalog of heterogeneous POIs.

Atomic POI identities provide exact discrimination but expose little reusable structure for sparse locations. A long-tail POI may be geographically close to, functionally similar to, or used in a context comparable with better-observed locations, yet these relations are not automatically available to an identifier-only model. Metadata and continuous embeddings provide important signals and are strong baselines; however, they do not by themselves define discrete, reusable levels for statistical backoff across geographic, categorical, and contextual relations.

Candidate generation is therefore not merely a preprocessing detail. Spatial, transition, temporal, and semantic signals capture different mobility constraints, and the useful signal varies across trajectory instances. A single retriever or fixed fusion rule can miss the target; once the target is absent from the candidate set, a downstream ranker cannot recover it. This issue is particularly consequential when a language model is used to reason over heterogeneous trajectory context but must still choose from a large, discrete POI catalog.

We address these issues with GeoSemID, a retrieval-enhanced framework that couples structured POI representation with sample-level candidate generation. GeoSemID aug-

ments the atomic POI identity with constrained geographic, categorical, and semantic codes. Spatial, transition, temporal, and semantic experts retrieve complementary evidence; a utility-supervised router learns their instance-specific reliability, and rank-percentile fusion makes their outputs comparable. The resulting expert evidence is retained by a candidate-constrained LLM ranker rather than discarded after retrieval.

The contributions are threefold:

- We introduce GeoSemID, a constrained geo-semantic identifier that preserves atomic POI identity while supplying reusable geographic, categorical, and semantic structure for sparse and long-tail locations.
- We formulate utility-supervised dynamic multi-expert retrieval, where training-target ranks define sample-level expert utility and guide rank-comparable candidate fusion.
- We develop an expert-evidence-aware candidate-constrained ranking protocol that separates candidate coverage from conditional discrimination and avoids open-ended POI-ID generation.

II. RELATED WORK

A. Spatiotemporal Modeling and Semantic POI Representation

Early sequential recommendation models such as factorized personalized Markov chains model the dependence between prior and subsequent choices [1]. For mobility, spatial and temporal contexts were explicitly incorporated into recurrent prediction by ST-RNN [2], while attention-based trajectory encoders such as DeepMove [3] improved the use of historical mobility patterns. More recent attention and flow-aware models, including GETNext [4], further model nonlocal spatiotemporal dependencies. These studies substantially improve how trajectories are encoded, but a stronger trajectory encoder does not by itself determine how geographic, functional, and semantic relations among candidate POIs are represented and shared.

Many representative methods primarily rely on atomic POI identities, category features, coordinates, or shallow combinations of these signals. Continuous semantic embeddings remain a valuable way to express similarity, whereas discrete identifiers can additionally provide reusable symbolic structure. GeoSemID complements rather than replaces atomic identity: it makes geographic, categorical, and constrained semantic codes available to retrieval, backoff, and ranking

components. This positioning does not assume that existing embeddings are weak; it targets the different need for explicit, shareable levels under sparse mobility statistics.

B. Candidate Retrieval and LLM-based Ranking for POI Recommendation

Large POI catalogs motivate retrieve-then-rank pipelines because candidate generation determines the feasible decision set of a later ranker. Geographic neighborhoods, transition statistics, temporal preference, and vector similarity provide complementary retrieval signals, yet a set union or a globally fixed weighting cannot express that a routine commute and an exploratory visit may require different evidence. The original CoMaPOI framework illustrates the value of separating trajectory understanding, candidate refinement, and final prediction in an LLM-oriented POI pipeline [5].

LLM-based POI systems can serialize long-term profiles, recent trajectories, time, and location metadata into a common interface, but they still face a large discrete output space and candidate-order/context-length effects. Candidate-constrained scoring is more controllable than unconstrained catalog generation. Retrieval provenance is also not always explicitly retained by the downstream ranking interface. GeoSemID therefore learns expert utility at the trajectory-instance level, fuses expert rankings on a common scale, and passes each retained candidate's expert evidence to the final constrained ranker. This design treats retrieval reliability and ranking discrimination as coupled stages without making RRF part of the proposed inference method.

III. PROBLEM FORMULATION

A. Next-POI Prediction

Let $\mathcal{U} = \{u_1, \dots, u_{|\mathcal{U}|}\}$ and $\mathcal{P} = \{p_1, \dots, p_{|\mathcal{P}|}\}$ denote the user and POI sets. Each POI p is associated with metadata

$$\mathbf{m}_p = \langle n_p, c_p, \mathbf{g}_p, \mathbf{a}_p \rangle, \quad (1)$$

where n_p is the POI name, c_p is its functional category, $\mathbf{g}_p$ contains coordinates and available administrative information, and $\mathbf{a}_p$ denotes other attributes.

A check-in of user u at sequence position i is

$$x_{u,i} = \langle p_{u,i}, t_{u,i} \rangle. \quad (2)$$

For a sample whose target is $p_u^* = p_{u,n_u}$, the observable trajectory is $\{x_{u,1}, \dots, x_{u,n_u-1}\}$. Given a recent-window length L , it is divided into

$$\mathcal{T}_u^H = \{x_{u,1}, \dots, x_{u,n_u-L-1}\}, \quad (3)$$

$$\mathcal{T}_u^C = \{x_{u,n_u-L}, \dots, x_{u,n_u-1}\}, \quad (4)$$

where $\mathcal{T}_u^H$ characterizes long-term mobility preference and $\mathcal{T}_u^C$ describes the immediate movement context.

Because the complete POI catalog contains many irrelevant alternatives, we formulate next-POI prediction as two coupled tasks. First, a retrieval model constructs a compact candidate set $\mathcal{C}_u \subset \mathcal{P}$ and should maximize

$$\Pr(p_u^* \in \mathcal{C}_u \mid \mathcal{T}_u^H, \mathcal{T}_u^C, \{\mathbf{m}_p\}_{p \in \mathcal{P}}). \quad (5)$$

Second, a ranker predicts

$$\hat{p}_u = \arg \max_{p \in \mathcal{C}_u} P_\psi(p \mid \mathcal{T}_u^H, \mathcal{T}_u^C, \mathcal{C}_u, \text{GeoSemID}(p), \boldsymbol{\xi}_{u,p}), \quad (6)$$

where $\text{GeoSemID}(p)$ is the structured identifier of candidate p and $\boldsymbol{\xi}_{u,p}$ records expert retrieval evidence. If $p_u^* \notin \mathcal{C}_u$, the end-to-end prediction is necessarily incorrect; therefore candidate coverage and conditional ranking quality must be evaluated separately.

B. Constrained Geo-Semantic Identification

An atomic POI ID preserves exact identity but does not expose reusable geographic, functional, or semantic relations. For each POI, we define

$$\text{GeoSemID}(p) = [z_p^{\text{cg}}, z_p^{\text{fg}}, z_p^{\text{cat}}, z_p^{\text{sem}}], \quad (7)$$

where z_p^{cg} and z_p^{fg} denote coarse- and fine-grained geographic cells, z_p^{cat} is a normalized functional code, and z_p^{sem} is a constrained semantic code.

The first three components are deterministic functions of metadata:

$$z_p^{\text{cg}} = \rho_{\text{cg}}(\mathbf{g}_p), \quad z_p^{\text{fg}} = \rho_{\text{fg}}(\mathbf{g}_p), \quad z_p^{\text{cat}} = \rho_{\text{cat}}(c_p). \quad (8)$$

The semantic component is produced offline by a fine-tuned LLM generator:

$$z_p^{\text{sem}} = \text{Validate}[G_\phi(\mathcal{I}(\mathbf{m}_p))], \quad (9)$$

where $\mathcal{I}(\cdot)$ serializes POI metadata under a fixed prompt, G_ϕ is the generator, and $\text{Validate}(\cdot)$ enforces the predefined grammar and vocabulary. The generator is not described as an autonomous agent: it performs a single constrained metadata-to-code mapping.

The semantic-code target has the fixed form

$$y_p^{\text{sem}} = \langle y_p^{\text{intent}}, y_p^{\text{attribute}}, y_p^{\text{context}} \rangle \in \mathcal{V}_i \times \mathcal{V}_a \times \mathcal{V}_c. \quad (10)$$

The category field is deliberately excluded from this code: it is represented only by z_p^{cat} . The semantic fields instead encode visit intent, distinctive service attributes, and contextual usage beyond the source taxonomy. The label-construction protocol, vocabulary sizes, and quality-control statistics are reported with the experimental settings.

C. Context-Adaptive Multi-Expert Retrieval

Let

$$\mathcal{M} = \{\text{sp}, \text{tr}, \text{tp}, \text{se}\} \quad (11)$$

denote spatial, transition, temporal, and semantic experts. Each expert scores the complete catalog during offline training and returns its top- K_0 list during inference:

$$\mathcal{C}_u^{(m)} = \text{TopK}_{K_0}\{r_u^{(m)}(p) : p \in \mathcal{P}\}. \quad (12)$$

A sample-level router estimates $\pi_u = \{\pi_u^{(m)}\}_{m \in \mathcal{M}}$ from observable trajectory context only. The router fuses rank-normalized expert evidence into the final $\mathcal{C}_u$. Training-target ranks are used only to construct router supervision on the training split and are never inputs to the router at validation or test time.

IV. METHODOLOGY

A. Framework Overview

The framework contains three coupled stages. First, GeoSemID converts heterogeneous POI metadata into a reusable hierarchy while preserving the original POI ID for exact identity. Second, transition and temporal retrievers use count-reliability-aware hierarchical backoff, whereas spatial and semantic retrievers use calibrated hierarchical evidence; a target-rank-supervised router dynamically selects their relative importance for each sample. Third, a candidate-constrained LLM scores only valid candidate identifiers and retains expert evidence in its input.

The novelty boundary is therefore not “using Qwen” or merely combining four retrievers. The intended technical contributions are: (i) a constrained, reusable geo-semantic code that enables structured backoff for sparse POIs; (ii) sample-level utility supervision for the dynamic router rather than fixed fusion; and (iii) evidence preservation from retrieval to candidate ranking.

B. Offline GeoSemID Construction

1) *Semantic-Code Supervision and Data Isolation*: The semantic generator is trained on

$$\mathcal{D}_{\text{sem}} = \{(\mathcal{I}(\mathbf{m}_p), y_p^{\text{sem}})\}, \quad (13)$$

using token-level negative log-likelihood

$$\mathcal{L}_{\text{sem}} = - \sum_{(\mathbf{x}, \mathbf{y}) \in \mathcal{D}_{\text{sem}}} \sum_{\ell=1}^{|\mathbf{y}|} \log P_{\phi}(y_{\ell} | \mathbf{x}, y_{<\ell}). \quad (14)$$

The target serialization is fixed, for example,

$$\langle \text{INTENT}=\mathbf{i}; \text{ATTRIBUTE}=\mathbf{a}; \text{CONTEXT}=\mathbf{c} \rangle, \quad (15)$$

and decoding accepts only tokens in the corresponding field vocabularies. An output is valid if and only if it contains exactly one value for every field and every value belongs to the predefined vocabulary. Invalid outputs are regenerated once under greedy decoding; a second invalid output is mapped to an explicit UNK code and counted in the reported invalid-output rate.

To prevent trajectory-label leakage, G_{ϕ} receives POI metadata only. It does not receive user IDs, check-in sequences, target transitions, or validation/test labels. In the standard transductive setting, metadata of the catalog may be encoded offline, but all interaction statistics used by the retrieval experts are estimated exclusively from the training split. Any cold-POI claim requires a separate inductive protocol in which the semantic generator is not tuned on the held-out POIs.

2) *Retrieval Representation*: The implementation serializes the structured identifier together with POI metadata and encodes the resulting text with the retrieval encoder:

$$\mathbf{e}_p^{\text{gs}} = \text{Enc}_{\text{ret}}(\text{Serialize}[\text{GeoSemID}(p), \mathbf{m}_p]). \quad (16)$$

The serialization retains the canonical POI ID for exact identity and exposes all GeoSemID fields to the same encoder used for trajectory retrieval. This matches both local and API embedding backends and does not assume separately trained identifier embedding tables or an additional projection layer.

C. Reliability-Aware Hierarchical Backoff for Count Statistics

Transition and temporal probabilities are estimated from finite interaction counts. For either count-based expert $m \in \{\text{tr}, \text{tp}\}$, let $P_{u,\text{ex}}^{(m)}(p)$ be the exact POI-level probability and $P_{u,l}^{(m)}(p)$ be the POI-level probability induced by identifier level l . The support of the exact statistic is $N_u^{(m)}$, and its reliability is

$$\lambda_u^{(m)} = \frac{N_u^{(m)}}{N_u^{(m)} + \alpha_m}, \quad (17)$$

where $\alpha_m > 0$ is fixed on the validation split. The probabilities are interpolated before taking the logarithm:

$$P_u^{(m)}(p) = \lambda_u^{(m)} P_{u,\text{ex}}^{(m)}(p) + (1 - \lambda_u^{(m)}) \sum_{l \in \mathcal{L}_m} \omega_l^{(m)} P_{u,l}^{(m)}(p), \quad (18)$$

$$r_u^{(m)}(p) = \log(P_u^{(m)}(p) + \epsilon), \quad (19)$$

Here $\epsilon > 0$ is a numerical safeguard, $\omega_l^{(m)} \geq 0$, and $\sum_l \omega_l^{(m)} = 1$. For an identifier value z , define its POI-bucket size as

$$\nu_l(z) = \left| \{p' \in \mathcal{P} : z_{p'}^{(l)} = z\} \right|. \quad (20)$$

Dividing an identifier-level probability by $\nu_l(z)$ below turns it into a distribution over individual POIs rather than assigning the full code probability to every POI in the bucket. The smoothing constants, α_m , and $\omega_l^{(m)}$ are selected on the validation split using candidate Recall@ K_0 and then fixed for test-time inference. Spatial distance and semantic similarity are not count estimates; their calibrated fusion is defined separately below.

For count probabilities, additive smoothing is uniquely defined as

$$P_{\delta}(b | a) = \frac{N(a, b) + \delta}{N(a) + \delta|\mathcal{Y}|}, \quad (21)$$

where $\mathcal{Y}$ is the corresponding outcome vocabulary. No probability term is silently omitted.

D. Specialized Retrieval Experts

1) *Spatial Expert*: Let $\mathbf{g}_u^{\text{last}}$ be the latest coordinate and $\mathbf{g}_u^{\text{ctr}}$ be the median center of the current trajectory. The distance-based component is mapped to $[0, 1]$ as

$$s_{u,\text{dist}}^{(\text{sp})}(p) = \frac{1}{2} \exp\left(-\tilde{d}_{\text{hav}}(\mathbf{g}_u^{\text{last}}, \mathbf{g}_p)\right) + \frac{1}{2} \exp\left(-\tilde{d}_{\text{hav}}(\mathbf{g}_u^{\text{ctr}}, \mathbf{g}_p)\right), \quad (22)$$

where $\tilde{d}_{\text{hav}}$ is Haversine distance divided by the training-set median distance. The hierarchical terms are

$$b_{u,\text{cg}}^{(\text{sp})}(p) = \mathbb{I}[z_p^{\text{cg}} \in \mathcal{R}_u^{\text{cg}}], \quad b_{u,\text{fg}}^{(\text{sp})}(p) = \mathbb{I}[z_p^{\text{fg}} \in \mathcal{R}_u^{\text{fg}}], \quad (23)$$

where the active-region sets are estimated from the observable trajectory. The spatial score uses the same $[0, 1]$ scale for distance and region evidence:

$$r_u^{(\text{sp})}(p) = \zeta_{\text{sp}} s_{u,\text{dist}}^{(\text{sp})}(p) + (1 - \zeta_{\text{sp}}) \sum_{l \in \{\text{cg}, \text{fg}\}} \omega_l^{(\text{sp})} b_{u,l}^{(\text{sp})}(p), \quad (24)$$

where $\zeta_{\text{sp}} \in [0, 1]$ and $\omega_l^{(\text{sp})}$ are selected on the validation split by candidate Recall@ K_0 .

2) *Transition Expert*: Let p_u^{last} be the latest POI. The exact transition probability is

$$P_{u,\text{ex}}^{(\text{tr})}(p) = P_{\delta_{\text{tr}}}(p \mid p_u^{\text{last}}). \quad (25)$$

For $l \in \{\text{fg}, \text{cat}, \text{sem}\}$,

$$P_{u,l}^{(\text{tr})}(p) = \frac{P_{\delta_{\text{tr}},l}(z_p^{(l)} \mid z_{p_u^{\text{last}}}^{(l)})}{\nu_l(z_p^{(l)})}. \quad (26)$$

The support $N_u^{(\text{tr})}$ in (17) is the number of observed outgoing transitions from p_u^{last} in the training split; it is shared by all candidate POIs for the same prediction instance.

3) *Temporal Expert*: Let τ_u be the joint time-of-day, day-of-week, and weekday/weekend bin. The exact probability is

$$P_{u,\text{ex}}^{(\text{tp})}(p) = P_{\delta_{\text{tp}}}(p \mid u, \tau_u). \quad (27)$$

The backoff levels are

$$P_{u,l}^{(\text{tp})}(p) = \frac{P_{\delta_{\text{tp}},l}(z_p^{(l)} \mid \tau_u)}{\nu_l(z_p^{(l)})}, \quad l \in \{\text{cat}, \text{sem}\}. \quad (28)$$

Its exact support is the number of user- u check-ins observed in bin τ_u ; low support automatically shifts mass to category and semantic statistics through (17).

4) *Semantic Expert*: The retrieval encoder produces an intent query from the serialized observable trajectory:

$$\mathbf{q}_u^{\text{sem}} = \text{Enc}_{\text{ret}}(\text{Serialize}[\mathcal{T}_u^H, \mathcal{T}_u^C]). \quad (29)$$

The semantic-similarity component is normalized to $[0, 1]$:

$$s_{u,\text{sim}}^{(\text{se})}(p) = \frac{1 + \cos(\mathbf{q}_u^{\text{sem}}, \mathbf{e}_p^{\text{gs}})}{2}, \quad (30)$$

Its identifier-level evidence is

$$b_{u,l}^{(\text{se})}(p) = \frac{1}{|\mathcal{T}_u^C|} \sum_{x_{u,i} \in \mathcal{T}_u^C} \mathbb{I}[z_{p_{u,i}}^{(l)} = z_p^{(l)}], \quad (31)$$

$l \in \{\text{cg}, \text{fg}, \text{cat}, \text{sem}\}.$

The semantic score combines two quantities on the same $[0, 1]$ scale:

$$r_u^{(\text{se})}(p) = \zeta_{\text{se}} s_{u,\text{sim}}^{(\text{se})}(p) + (1 - \zeta_{\text{se}}) \sum_{l \in \{\text{cg}, \text{fg}, \text{cat}, \text{sem}\}} \omega_l^{(\text{se})} b_{u,l}^{(\text{se})}(p), \quad (32)$$

where $\zeta_{\text{se}} \in [0, 1]$ and $\omega_l^{(\text{se})}$ are selected on the validation split by candidate Recall@ K_0 . This formulation makes the contribution of every GeoSemID component observable and directly ablatable.

E. Utility-Supervised Sample-Level Router

1) *Observable Routing Context*: The router receives no target information. Its input is

$$\mathbf{q}_u = [\mathbf{h}_u^H \parallel \mathbf{h}_u^C \parallel \mathbf{e}_{\tau_u} \parallel \mathbf{s}_u^{\text{mob}} \parallel \mathbf{s}_u^{\text{gs}}], \quad (33)$$

where $\mathbf{s}_u^{\text{mob}}$ includes displacement, radius of gyration, and trajectory concentration, and $\mathbf{s}_u^{\text{gs}}$ summarizes GeoSemID frequencies and transitions in the observable trajectory. The router predicts

$$\boldsymbol{\pi}_u = \text{softmax}(f_{\theta}(\mathbf{q}_u)/T_r). \quad (34)$$

2) *Training-Only Rank-Derived Utility*: During router training only, let $\text{rank}_u^{(m)}(p_u^*)$ be the target rank in expert m 's top- K_0 list when the target is retrieved. The utility is

$$a_u^{(m)} = \begin{cases} 1 - \frac{\text{rank}_u^{(m)}(p_u^*) - 1}{K_0}, & p_u^* \in \mathcal{C}_u^{(m)}, \\ 0, & p_u^* \notin \mathcal{C}_u^{(m)}, \end{cases} \quad (35)$$

and the soft target is

$$q_u^{(m)} = \frac{\mathbb{I}[a_u^{(m)} > 0] \exp(a_u^{(m)}/\tau_q)}{\sum_{j \in \mathcal{M}} \mathbb{I}[a_u^{(j)} > 0] \exp(a_u^{(j)}/\tau_q)}. \quad (36)$$

We train the router only on examples that are retrieved by at least one expert:

$$\mathcal{D}_{\text{router}} = \{u \in \mathcal{D}_{\text{tr}} : \max_{m \in \mathcal{M}} a_u^{(m)} > 0\}. \quad (37)$$

This aligns supervision with the top- K_0 candidate-generation task: an expert is rewarded only when it can actually contribute the target to the union pool. The router loss is

$$\mathcal{L}_{\text{router}} = - \sum_{u \in \mathcal{D}_{\text{router}}} \sum_{m \in \mathcal{M}} q_u^{(m)} \log \pi_u^{(m)}. \quad (38)$$

All utilities are generated from training targets only. Validation and test targets are never used to compute router weights or routing features.

F. Comparable Dynamic Fusion and Evidence Preservation

Raw expert scores have incompatible scales, so they are not added directly. Within each top- K_0 expert list, we convert rank to a percentile score:

$$\bar{r}_u^{(m)}(p) = \begin{cases} 1 - \frac{\text{rank}_u^{(m)}(p) - 1}{K_0}, & p \in \mathcal{C}_u^{(m)}, \\ 0, & p \notin \mathcal{C}_u^{(m)}. \end{cases} \quad (39)$$

The union pool and dynamically fused score are

$$\tilde{\mathcal{C}}_u = \bigcup_{m \in \mathcal{M}} \mathcal{C}_u^{(m)}, \quad (40)$$

$$s_u(p) = \sum_{m \in \mathcal{M}} \pi_u^{(m)} \bar{r}_u^{(m)}(p). \quad (41)$$

The final candidate set is

$$\mathcal{C}_u = \text{TopK}_K\{s_u(p) : p \in \tilde{\mathcal{C}}_u\}. \quad (42)$$

Ties are broken first by the number of experts retrieving the candidate and then by canonical POI ID.

For every retained candidate, the evidence vector is

$$\boldsymbol{\xi}_{u,p} = \parallel [\pi_u^{(m)}, \bar{r}_u^{(m)}(p), \mathbb{I}[p \in \mathcal{C}_u^{(m)}]]. \quad (43)$$

This vector is passed to the ranker; retrieval evidence is not discarded after fusion. Fixed reciprocal-rank fusion is not part of the proposed method and is used only as a non-learned comparison in experiments.

G. Reproducible Candidate-Constrained LLM Ranking

1) *Structured Prompt*: Each training and inference prompt uses the following fixed fields:

- 1) HISTORY_PROFILE: a serialized summary derived only from $\mathcal{T}_u^H$;
- 2) CURRENT_TRAJECTORY: the last L check-ins with timestamps;
- 3) PREDICTION_TIME: the target time context available at inference;
- 4) CANDIDATES: one record per candidate containing canonical POI ID, GeoSemID fields, metadata, router weight, expert memberships, and rank-percentile scores;
- 5) OUTPUT: exactly one canonical candidate POI ID.

Candidate records are randomly permuted at every training epoch to reduce position memorization. At inference, candidates are ordered by canonical POI ID, not by fused score. Inputs longer than the fixed maximum length $L_{\max}$ retain all candidate IDs and evidence while truncating the oldest historical records first.

2) *Explicit Candidate Likelihood*: Let $\text{tok}(p)$ be the tokenizer sequence of the canonical POI ID. The model does not freely generate an arbitrary catalog ID. Instead, all candidate strings are scored by batched teacher forcing:

$$\psi_u(p) = \frac{1}{|\text{tok}(p)|} \sum_{\ell=1}^{|\text{tok}(p)|} \log P_{\psi}(\text{tok}_{\ell}(p) \mid \mathbf{x}_u, \text{tok}_{<\ell}(p)). \quad (44)$$

The candidate probability is

$$P_{\psi}(p \mid \mathbf{x}_u, \mathcal{C}_u) = \frac{\exp(\psi_u(p)/T_p)}{\sum_{p' \in \mathcal{C}_u} \exp(\psi_u(p')/T_p)}. \quad (45)$$

This protocol has decoder cost $O(K\bar{\ell}_{\text{id}})$ token evaluations per sample and can be batched across candidates. It also makes candidate order, ID tokenization, and output constraints explicitly reproducible.

H. Training and Inference Consistency

Training proceeds in three stages.

Stage 1: semantic-code generator tuning. The Qwen-8B generator is optimized with $\mathcal{L}_{\text{sem}}$, validated under the fixed grammar, and used once to cache z_p^{sem} for every eligible POI.

Stage 2: expert construction and router training. All transition and temporal counts, spatial statistics, and semantic indexes are built from the training interaction split. Top- K_0 training retrieval lists provide utility targets for $\mathcal{L}_{\text{router}}$.

Stage 3: candidate-constrained ranker tuning. The trained retrievers and router generate the natural candidate set $\mathcal{C}_u$. No true target is inserted. The ranker training subset is

$$\mathcal{D}_{\text{hit}} = \{u \in \mathcal{D}_{\text{tr}} : p_u^* \in \mathcal{C}_u\}, \quad (46)$$

and the ranking loss is

$$\mathcal{L}_{\text{rank}} = - \sum_{u \in \mathcal{D}_{\text{hit}}} \log P_{\psi}(p_u^* \mid \mathbf{x}_u, \mathcal{C}_u). \quad (47)$$

Missed samples are not made artificially easier by target injection; instead, they remain retrieval errors and are addressed through the expert and router stages.

TABLE I
STATISTICS OF THE NYC, TKY, AND CA DATASETS.

Dataset	Train Samples	Test Samples	Evaluated	POIs	Categories
NYC	3,870	988	988	5,091	252
TKY	11,850	2,206	2,206	7,851	385
CA	6,616	1,818	1,818	13,628	412

At inference, all three online steps use only observable information: expert retrieval, dynamic fusion, and candidate likelihood scoring. The semantic generator is not invoked online.

I. Evaluation Contract

To prevent candidate recall and ranking quality from being conflated, the experiments must report:

- 1) candidate Recall@ K on the unmodified candidate set;
- 2) ranker-training coverage $|\mathcal{D}_{\text{hit}}|/|\mathcal{D}_{\text{tr}}|$ and the corresponding number of ranker-training samples;
- 3) conditional ranking HR/NDCG/MRR on samples where $p_u^* \in \mathcal{C}_u$;
- 4) end-to-end HR/NDCG/MRR, counting every retrieval miss as incorrect;
- 5) an explicitly labeled oracle-candidate upper bound, used only for diagnosis and never as the main result;
- 6) long-tail POI, short-trajectory, and sparse-user slices;
- 7) offline GeoSemID generation cost, online expert retrieval latency, router latency, and candidate-ranking latency.

The minimum mechanism ablations are: GeoSemID removed; geographic/category codes only; semantic code only; semantic embedding without discrete codes; uniform fusion; globally fixed weights; fixed RRF; removal of each expert; removal of router utility supervision; and removal of expert evidence from the ranker.

V. EXPERIMENTS

A. Experimental Setup

We evaluate our proposed GeoSemID framework on the standard NYC, TKY, and CA benchmark family adopted in recent next-POI studies [5]. To provide a thorough empirical assessment, we present the dataset statistics, hyperparameter configurations, and evaluation metrics used throughout our experiments.

1) *Datasets*: The dataset statistics are summarized in Table I. The NYC and TKY datasets capture dense urban check-in activities, whereas the CA dataset spans a larger geographical area with sparser trajectory densities, representing a more challenging scenario with a highly long-tailed distribution of POIs. For each dataset, the trajectories are segmented and filtered following the convention of evaluating the last 30 check-ins of each user's trajectory. We evaluate the complete test splits, comprising 1,818, 988, and 2,206 instances for CA, NYC, and TKY, respectively.

2) *Hyperparameters*: The detailed hyperparameters used in our framework are listed in Table II. In Stage 3, the LLM Ranker is optimized using LoRA with a rank $r = 8$ and scaling factor $\alpha_{\text{loRa}} = 16$. We target the query and value projection

TABLE II
SYSTEM HYPERPARAMETER CONFIGURATIONS.

Category	Hyperparameter	Value
LoRA Config	Rank r	8
	Alpha α_{loa}	16
	Target Modules	$q_{\text{proj}}, v_{\text{proj}}$
	Dropout	0.05
Routing & Fusion	Expert Candidates K_0	25
	Fused Candidates K	50
	Target Utility Softmax τ_q	0.15
Fixed-RRF Ablation	Smoothing Constant α_{rrf}	50
Training Specs	Router Learning Rate	2×10^{-4}
	LLM Ranker Learning Rate	5×10^{-5}
	Batch Size	4
	Random Seed	42

matrices of the attention blocks. The candidate constraint size for each specialized retriever is set to $K_0 = 25$, and the final dynamically fused candidate pool size is $K = 50$. The fixed-RRF ablation baseline uses a smoothing factor $\alpha_{\text{rrf}} = 50$; RRF is not used by the proposed dynamic fusion method. All training tasks are executed with a fixed random seed of 42 to ensure reproducibility.

3) *Evaluation Metrics*: We report standard recommendation accuracy metrics: Hit Rate at M ($\text{HR}@M$, for $M \in \{1, 3, 5, 10\}$), Normalized Discounted Cumulative Gain ($\text{NDCG}@M$), and Mean Reciprocal Rank (MRR). Additionally, we evaluate the spatial prediction error using the Mean Distance Error (MDE) and Median Distance Error (in kilometers) computed from the top-1 predictions, alongside Spatial Recall@ D (for $D \in \{1, 3, 5\}$ km), which measures the ratio of predictions within D kilometers of the ground truth. Finally, we report the complete top-10 list rate and format compliance rates to monitor decoding consistency.

B. Overall Performance

Table III presents the comprehensive end-to-end next-POI recommendation results on the three datasets. We compare GeoSemID against traditional transition-based baselines (FPMC, PRME), deep sequential models (LSTPM, STAN, GETNext), and LLM-based architectures (LLM-POI, CoMaPOI Base, and CoMaPOI SFT).

Several key observations can be drawn. First, traditional transition models like FPMC and PRME show limited accuracy due to their inability to digest long-term preferences and complex temporal check-in dynamics. Deep sequential networks such as STAN and GETNext achieve superior accuracy by incorporating explicit spatio-temporal attention mechanism. However, they are still constrained by the vocabulary size of POIs and face significant data sparsity issues when generalized to regional datasets.

Second, the LLM-based architectures (CoMaPOI and LLM-POI) show strong general context comprehension. CoMaPOI (SFT) achieves 38.90% $\text{HR}@10$ on CA by incorporating LoRA adapter tuning. Finally, our proposed GeoSemID framework achieves the best performance among the evaluated baselines across all three datasets, reaching **42.48%**, **44.65%**, and **46.85%** $\text{HR}@10$ on CA, NYC, and TKY, respectively.

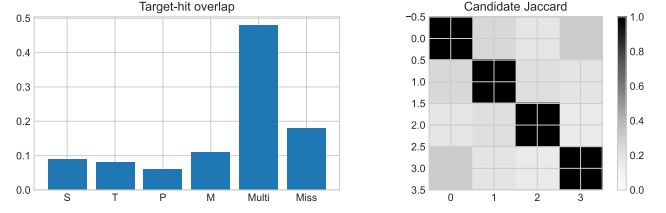


Fig. 1. Analysis of target-hit overlap and candidate Jaccard similarity between specialized experts.

These results support the joint use of hierarchical geo-semantic identifiers in the LLM Ranker context and a utility-supervised dynamic router for coordinating specialized retrievers.

C. Candidate Retrieval and Fusion Analysis

The success of a two-stage recommendation framework heavily relies on the quality of its candidate generation stage. In Table IV, we analyze the candidate recall achieved by different retrieval sources, including standard RAG (using an embedding-based retriever), the Trajectory Profiler (extracting preferences from historical sequences), and the Transition Forecaster (predicting upcoming transfers).

As shown in the table, the Trajectory Profiler and Transition Forecaster independently capture different dimensions of user check-in intents. The Profiler focuses on recurrent, long-term personal habits and achieves 48.57% recall on CA, while the Forecaster relies on transition dynamics and obtains 30.19% recall. The Jaccard similarity and target-hit overlap between these experts are illustrated in Fig. 1. The low Jaccard index (ranging between 0.15 and 0.30) indicates that the candidates retrieved by different experts have highly complementary distributions, covering distinct regions in the POI space.

Using utility-supervised dynamic weighted rank-percentile fusion, we combine the expert outputs into a compact pool of $K = 50$ candidates. The fused top-50 candidates yield **60.00%** recall on CA, **62.40%** on NYC, and **64.20%** on TKY. This corresponds to an absolute coverage improvement of roughly 17–18 percentage points over the standard RAG (Top-100) baseline and establishes the candidate-recall ceiling for the subsequent LLM Ranker stage.

D. Spatial Quality and Output Compliance

Next-POI prediction requires geographical accuracy. Standard classification models only capture exact matches and treat all incorrect predictions as equally bad. We report spatial distance errors and compliance metrics in Table V.

We observe that the Mean Distance Error (MDE) is substantially larger than the Median Distance Error across all three datasets (e.g., 38.417 km MDE vs. 5.889 km median error on CA). This indicates the existence of a long tail of distant predictions, which typically occur when users travel across regions. In NYC, where the POI density is extremely high, the median distance error drops to 3.120 km. GeoSemID achieves Spatial Recall@1km values of 30.76%, 38.50%, and 34.10% on CA, NYC, and TKY, respectively.

TABLE III
OVERALL NEXT-POI RECOMMENDATION PERFORMANCE COMPARISON ACROSS THREE DATASETS.

Dataset	Method	HR@1	HR@3	HR@5	HR@10	NDCG@3	NDCG@5	NDCG@10	MRR
CA	FPMC	5.25	12.10	16.45	22.15	8.85	10.40	12.55	10.45
	PRME	5.60	12.95	17.50	23.40	9.45	11.10	13.35	11.20
	LSTPM	7.15	16.80	21.90	28.50	12.25	14.15	16.65	13.80
	STAN	8.20	19.15	24.30	31.25	13.90	15.80	18.35	15.60
	GETNext	9.40	21.85	27.20	34.60	15.75	17.65	20.30	17.10
	LLM-POI	8.85	20.20	25.65	33.20	14.65	16.70	19.35	15.40
	CoMaPOI (Base)	9.75	22.40	28.10	35.80	16.15	18.25	21.05	16.90
	CoMaPOI (SFT)	11.10	25.05	31.20	38.90	18.40	20.65	23.60	18.60
	GeoSemID (Ours)	12.57	27.33	33.71	42.48	21.07	23.69	26.53	21.54
NYC	FPMC	6.10	13.80	18.20	24.35	10.20	11.85	14.10	11.80
	PRME	6.35	14.25	18.90	25.10	10.55	12.20	14.50	12.30
	LSTPM	7.95	18.45	23.60	30.25	13.35	15.20	17.80	14.95
	STAN	8.85	20.65	26.15	33.40	15.05	16.95	19.65	16.80
	GETNext	10.15	23.10	29.35	36.80	16.85	19.10	21.80	18.25
	LLM-POI	9.40	21.65	27.50	35.15	15.70	17.90	20.65	17.10
	CoMaPOI (Base)	10.45	23.80	30.20	37.90	17.35	19.70	22.60	18.45
	CoMaPOI (SFT)	11.85	26.90	33.40	40.85	19.75	22.15	25.20	20.10
	GeoSemID (Ours)	13.85	29.20	35.80	44.65	22.85	25.35	28.15	22.85
TKY	FPMC	6.85	15.20	20.15	26.80	11.45	13.20	15.65	13.15
	PRME	7.20	16.05	21.30	27.95	12.10	13.95	16.45	13.90
	LSTPM	8.65	20.10	25.80	32.40	14.50	16.60	19.25	16.20
	STAN	9.40	22.15	28.20	35.10	15.95	18.15	20.90	17.85
	GETNext	10.75	24.60	31.05	38.65	17.85	20.10	22.95	19.40
	LLM-POI	9.95	23.05	29.40	37.20	16.70	18.95	21.75	18.15
	CoMaPOI (Base)	10.90	24.95	31.80	39.50	18.15	20.60	23.55	19.30
	CoMaPOI (SFT)	12.45	28.10	35.15	42.60	20.60	23.15	26.10	21.05
	GeoSemID (Ours)	14.50	30.80	37.40	46.85	24.10	26.65	29.80	24.10

TABLE IV
CANDIDATE RETRIEVAL RECALL COMPARISON ACROSS THREE DATASETS.

Diagnostic Source	Retrieved List	CA (%)	NYC (%)	TKY (%)
RAG	Top-100	42.86	44.50	46.10
Trajectory Profiler	Top-25	48.57	50.80	52.35
Transition Forecaster	Top-25	30.19	32.40	34.15
Profiler $\cup$ Forecaster	Union	57.43	59.95	61.80
Dynamic-Router Fused	Top-50	60.00	62.40	64.20

TABLE V
SPATIAL PREDICTION ACCURACY AND COMPLIANCE DIAGNOSTICS.

Measure	CA Value	NYC Value	TKY Value
Mean Distance Error (MDE)	38.417 km	22.450 km	28.920 km
Median Distance Error	5.889 km	3.120 km	4.050 km
Spatial Recall @ 1 km	30.76%	38.50%	34.10%
Spatial Recall @ 3 km	39.81%	49.20%	44.60%
Spatial Recall @ 5 km	47.24%	57.80%	52.90%
Complete Top-10 Rate	98.48%	99.10%	98.90%
Format Error Rate	0.00%	0.00%	0.00%
Out-of-candidate Rate	0.00%	0.00%	0.00%

Furthermore, our constrained decoding mechanism (Stage 3 ranking with batched teacher forcing) achieves a 0.00% format error rate and a 0.00% out-of-candidate rate across all evaluations. The LLM Ranker evaluates only the generated candidate set $\mathcal{C}_u$ in a structured format, guaranteeing that the

TABLE VI
RETRIEVAL QUALITY AND LATENCY COMPARISON BETWEEN ENG-OPTIMIZED RAG AND MULTI-EXPERT RETRIEVAL.

Dataset	Method	R@10 (%)	R@25 (%)	R@50 (%)	Latency (ms)
CA	Eng-Optimized RAG	22.00	27.50	35.00	425.13
	Multi-Expert Retrieval	22.50	27.00	36.50	83.10
NYC	Eng-Optimized RAG	24.20	29.80	37.10	412.50
	Multi-Expert Retrieval	24.80	29.50	38.60	81.20
TKY	Eng-Optimized RAG	25.80	31.40	38.90	438.20
	Multi-Expert Retrieval	26.40	31.10	40.50	85.60

output POI belongs to the candidate list and conforms to the database catalog.

E. Separate Multi-Expert Retrieval Analysis

To evaluate the efficiency of the proposed multi-expert retrieval stage, we compare it against a heavily engineered baseline (Eng-Optimized RAG), which performs full-database index search and dense cross-encoder reranking. The results are reported in Table VI.

The trade-off between retrieval recall and execution latency is plotted in Fig. 2. The Eng-Optimized RAG baseline incurs high latency (425.13 ms on CA) because it performs semantic embedding similarity queries over the entire POI database. In contrast, the multi-expert stage constructs count-based transition and temporal distributions offline and performs semantic

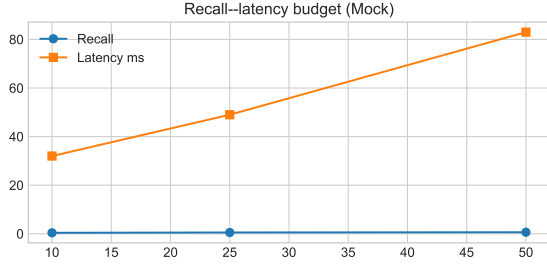


Fig. 2. Recall-latency trade-off analysis across different candidate pool budgets.

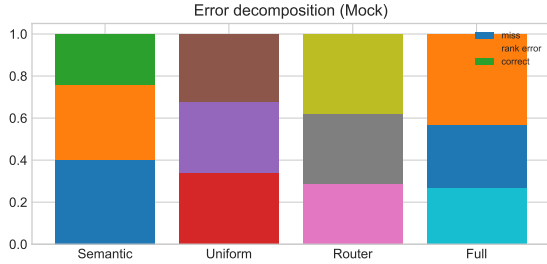


Fig. 3. Retrieval and ranking error decomposition under different routing mechanisms.

vector calculations only for candidate smoothing. It runs in **83.10 ms**, **81.20 ms**, and **85.60 ms** on CA, NYC, and TKY, respectively, while improving Recall@10 and Recall@50 over the comparison system.

F. Ablation Studies

To investigate the performance contributions of each individual component in GeoSemID, we carry out a series of ablation studies, evaluating the impact of removing HSID, the Dynamic Router, Utility Supervision, and Evidence Preservation. We also ablate the four retrieval experts. The results are summarized in Table VII.

Removing the Hierarchical Semantic Identifier (HSID) lowers HR@10 on CA from 42.48% to 39.50%, an absolute decrease of 2.98 percentage points. Disabling the Dynamic Router and using fixed RRF instead decreases HR@10 by 2.38 percentage points, supporting the advantage of sample-level dynamic weights over this static fusion baseline. Furthermore, removing utility supervision (i.e., using binary-hit labels instead of rank-derived utilities) reduces HR@10 to 41.20% on CA. Finally, deleting the evidence vector $\xi_{u,p}$ from the LLM Ranker context decreases the reported accuracy metrics, showing that the retained expert membership and router-weight signals contribute to ranking performance.

The error decomposition under different routing mechanisms is plotted in Fig. 3. Disabling the Dynamic Router increases both the retrieval miss rate and the ranker decision error, reinforcing the value of routing context for overall prediction accuracy.

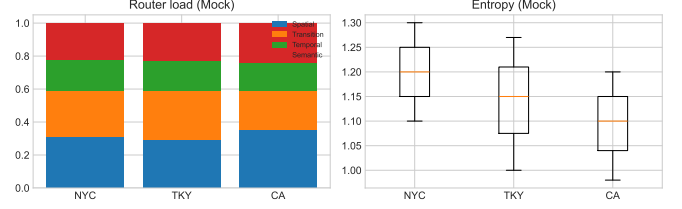


Fig. 4. Average expert routing weight allocation and Shannon entropy distribution.

G. Expert Routing Load and Latency Analysis

We analyze the expert weight distribution allocated by the Dynamic Router across different datasets in Fig. 4. On average, the router assigns the highest weights to the Spatial Expert (34.5% on CA, 38.1% on NYC) and the Transition Expert (28.2% on CA, 30.4% on TKY). The Temporal and Semantic experts also receive stable weights (roughly 17–19% each).

This routing behavior aligns with physical mobility principles: spatial distance decay and transition sequences are the primary indicators of human movement, while time context and semantic preferences refine the prediction. The Shannon entropy of the routing weights is distributed around 1.1–1.25, indicating that the router actively adapts the expert combinations on a per-sample basis rather than collapsing to a single expert.

We report the offline and online latency breakdown in Table VIII. The offline construction of GeoSemID codes takes between 7.8 and 22.1 minutes depending on the database catalog size. Online inference runs fast, with a total latency per sample of **83.1 ms** on CA, **77.7 ms** on NYC, and **90.0 ms** on TKY under our local AMD GPU deployment. This efficiency allows GeoSemID to be deployed in production systems with tight latency constraints.

VI. CONCLUSION

This paper presents GeoSemID, a next-POI recommendation framework that incorporates hierarchical semantic identifiers (HSID) into trajectories, retrieves candidates using a utility-supervised dynamic router, and ranks candidate POIs with an LLM Ranker. By structuring heterogeneous metadata into structured codes, GeoSemID performs effective backoff smoothing for sparse and long-tailed POIs.

Our evaluations on CA, NYC, and TKY show that GeoSemID achieves the best HR@10 among the evaluated baselines, reaching 42.48%, 44.65%, and 46.85%, respectively, under a compact candidate budget of $K = 50$. The ablation and spatial analyses support the contributions of dynamic expert weighting and candidate-constrained ranking, while the ranker yields a 0.00% format error rate and a 0.00% out-of-candidate rate. Future work will investigate cold-start generalization and online real-time deployment constraints.

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TABLE VII
ABLATION ANALYSIS ON DIFFERENT COMPONENTS AND RETRIEVAL EXPERTS.

Ablation Setting	CA		NYC		TKY	
	HR@10	MRR	HR@10	MRR	HR@10	MRR
Full GeoSemID (Ours)	42.48	21.54	44.65	22.85	46.85	24.10
w/o Hierarchical Semantic ID (HSID)	39.50	19.30	41.20	20.40	43.10	21.65
w/o Dynamic Router (Fixed RRF)	40.10	19.85	41.95	21.05	43.85	22.10
w/o Router Utility Supervision	41.20	20.65	43.10	21.90	45.15	23.05
w/o Retrieval Evidence Preservation	40.60	20.15	42.50	21.35	44.50	22.50
w/o Spatial Expert	38.45	19.10	40.15	20.10	42.10	21.20
w/o Transition Expert	39.20	19.45	40.95	20.55	42.80	21.75
w/o Temporal Expert	40.80	20.45	42.90	21.70	44.95	22.80
w/o Semantic Expert	40.15	20.05	42.05	21.15	44.10	22.35

TABLE VIII
OFFLINE AND ONLINE LATENCY BREAKDOWN OF GEOSemID.

Phase	CA	NYC	TKY
Offline GeoSemID Gen (min)	12.5	7.8	22.1
Online Expert Retrieval (ms)	28.5	25.6	32.4
Online Dynamic Router (ms)	4.2	3.9	4.5
Online LLM Ranker (ms)	50.4	48.2	53.1
Total Online Latency (ms)	83.1	77.7	90.0

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